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U–Pb geochronology of the Western Channel Diabase, northwestern Laurentia: Implications for a large 1.59 Ga magmatic province, Laurentia's APWP and paleocontinental reconstructions of Laurentia, Baltica and Gawler craton of southern Australia

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ABSTRACT

U–Pb baddeleyite ages of 1592 ± 3 and 1590 ± 4 Ma are reported for paleomagnetic sites in sheets and dykes of Western Channel Diabase (WCD) that intrude Proterozoic rocks of the flat-lying Hornby Bay Group in the Hornby Bay basin and the deformed volcanic-plutonic Great Bear Magmatic Zone of Wopmay orogen of northwestern Laurentia. A published WCD paleomagnetic pole at 9°N, 115°W (A_{95} = 6°) has been demonstrated primary. The new ages indicate that the WCD pole falls midway in time between poles for the 1.74 Ga Cleaver dykes and 1.48–1.42 Ga Elsonian-aged plutons, filling an important gap in the Proterozoic apparent polar wander path (APWP) for Laurentia. The WCD pole can be compared with poles reported from similar-aged magmatic units on other cratons in order to test paleocontinental reconstructions. A comparison of the Laurentian WCD pole with primary ca. 1.63 Ga and ca. 1.575 Ga poles for Baltica, along with an earlier comparison of precisely dated 1.27–1.255 Ga poles for Laurentia and Baltica, suggests that the two cratonic blocks drifted as a single entity with Baltica adjacent to eastern Greenland during the ca. 1.59–1.27 Ga interval. On the basis of less well constrained ca. 1.84–1.83 Ga poles from Laurentia and Baltica, it is possible that this reconstruction existed as early as ca. 1.83 Ga. The WCD is the same age as Wernecke breccias of the Wernecke and Ogilvie Mountains of northwestern Laurentia and bimodal Gawler Range Volcanics (GRV) and related Olympic Dam breccias of the Gawler craton. It has been proposed by others that the Gawler craton lay adjacent to northwestern Laurentia at 1.59 Ga, with the Olympic Dam and Wernecke breccias forming a large hydrothermal province. The primary WCD pole provides an opportunity to test Laurentia–Gawler craton reconstructions at 1.59 Ga. A paleopole has been reported for the GRV, although its primary or secondary nature is open to interpretation. If primary, or if acquired as an overprint during the later stages of 1.60–1.58 Ga Hiltaba-GRV magmatism, then a position for the Gawler craton adjacent to northwestern Laurentia is permitted. If the GRV pole is a later secondary overprint then a reliable comparison with Laurentian poles cannot be made.

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1. Introduction

Although Laurentia plays a pivotal role in many post-1.8 Ga paleocontinental reconstructions, such reconstructions for the latest Paleoproterozoic and earliest Mesoproterozoic are uncertain because of a lack of well-defined and well-dated paleomagnetic poles for this period. A potentially important paleopole for helping to resolve reconstructions at this time is that of the Western Channel Diabase (WCD) of northern Wopmay orogen and the adjacent Hornby Bay basin (Fig. 1). A magnetic remanence from the WCD was reported by Irving et al. (1972). However, prior to this study, the WCD was poorly dated, and the significance of its paleomagnetic remanence not known.

In this study, U–Pb baddeleyite ages are reported for two WCD samples collected in an unpublished paleomagnetic study by W.F. Fahrig and archived at the Geological Survey of Canada in Ottawa. Implications of the WCD age for the apparent polar wander path (APWP) of Laurentia, and for the reconstruction of Laurentia with other continental blocks, namely Baltica and the Gawler craton of southern Australia, are considered.

2. Geological setting

The Western Channel Diabase is widely distributed near eastern Great Bear Lake (Fig. 1b), Northwest Territories, Canada. In the vicinity of eastern Hornby Bay, Western Channel and Echo Bay

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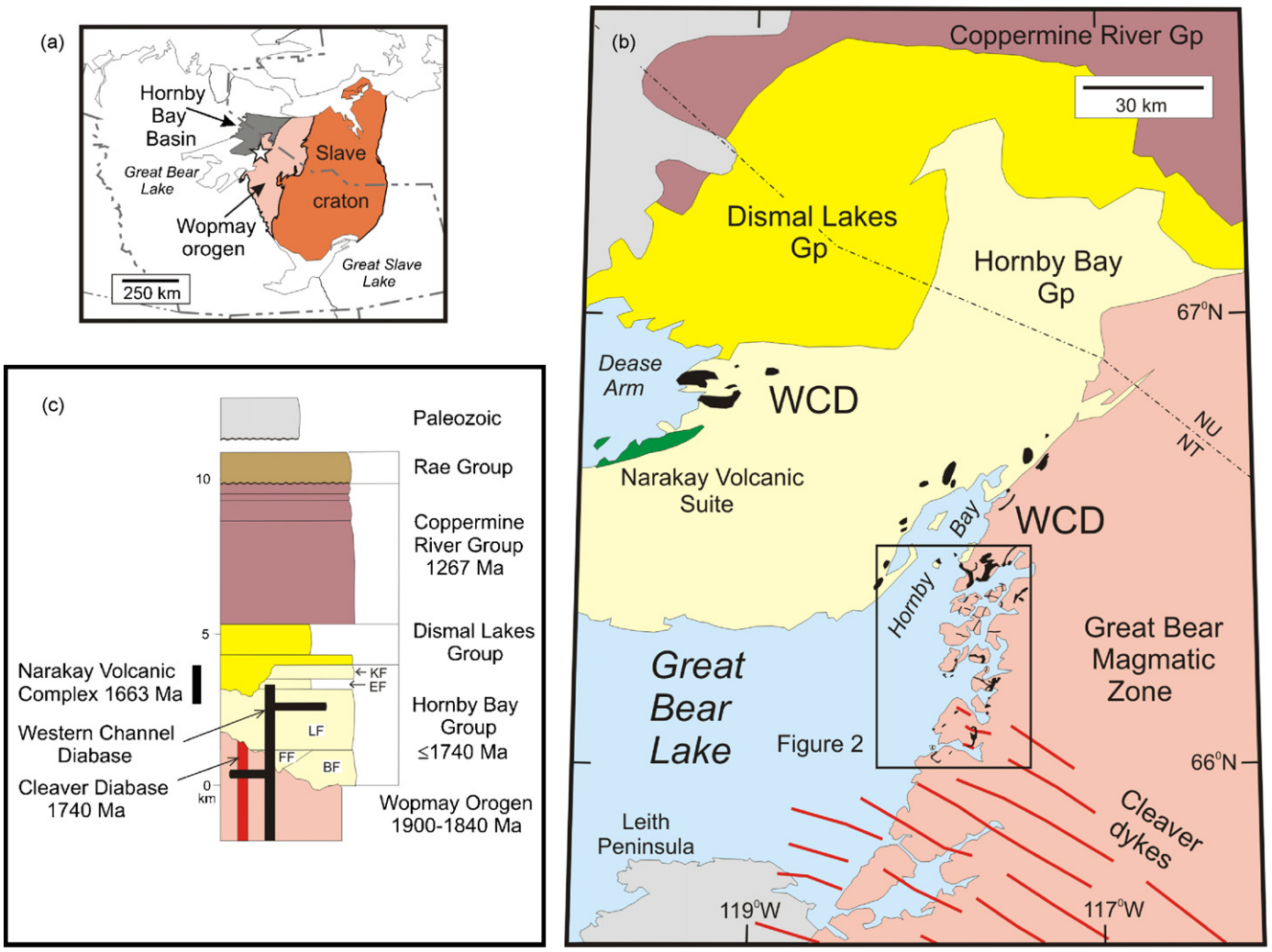


Fig. 1. (a) Location of study area (star) in northwestern Laurentia. (b) Geology in the vicinity of northeastern Great Bear Lake (modified after Hoffman and Hall, 1993; Ross and Kerans, 1989; Hildebrand, 1982; Feniak, 1952). Diabase sheets and dykes, mainly of Western Channel Diabase (WCD) shown in black. NU: Nunavut; NT: Northwest Territories. (c) Stratigraphy of the Hornby Bay basin (modified after Bowring and Ross, 1985). BF: Bigbear Formation; EF: East River Formation; FF: Fault River Formation; KF: Kaertok Formation; LF: Lady Nye Formation.

(Fig. 2), it occurs as sheets, sills and dykes that intrude the ca. 1890–1840 Ma Great Bear Magmatic Zone of Wopmay orogen and cuts 1740 +5/−4 Ma Cleaver dykes (Hildebrand, 1982; Irving et al., 2004). Immediately to the north it occurs as sheets within the overlying sedimentary rocks of the undeformed Hornby Bay Group in the Hornby Bay basin. Sheets also intrude strata of the Hornby Bay Group near Dease Arm (Ross and Kerans, 1989).

To date, the age of emplacement and crystallization of the WCD has been poorly constrained. A K–Ar age of 1405 ± 75 Ma (recalculated using the standardised tables of Harland et al., 1989) was reported by Wanless et al. (1970), and a Rb–Sr age of 1392 ± 48 Ma (recalculated using $\lambda^{87}\mathrm{Rb} = 1.42 \times 10^{-11}\ \mathrm{y}^{-1}$) was determined by Wanless et al. (1978). Because the WCD crosscuts the Cleaver dykes as noted above, the former must be younger than 1740 Ma. The WCD sheets have been observed to occur in the Hornby Bay Group up to the lower member of East River Formation (unit He1; Ross and Kerans, 1989), recently shown to contain detrital zircons as young as 1660–1620 Ma (Rainbird et al., 2009). This implies that the WCD can be no older than approximately 1620 Ma. Furthermore, the Narakay Volcanic Complex (Fig. 1; Ross, 1982), which contains bimodal volcanic rocks, also comprises shallow-marine sediments considered to be time-equivalent to the Lady Nye, East River and Kaertok formations of the upper Hornby Bay Group. Felsic volcanics in this complex have been dated at 1663 ± 8 Ma (Bowring and Ross, 1985). This age places an additional maximum age constraint on the WCD.

3. Previous paleomagnetic results

Irving et al. (1972) reported stable paleomagnetic remanences at 35 WCD sampling sites (Fig. 3a). The mean remanence direction is $D = 356°$, $I = -50°$ ($\alpha_{95} = 6°$, $k = 35$), with dyke sites yielding a significantly shallower mean direction than sheets (Fig. 3b). The paleopole corresponding to the overall WCD mean direction is 9°N, 115°W ($A_{95} = 6°$).

The number of separate intrusions that were sampled by Irving et al. (1972) is uncertain. They noted that six stably magnetized sites appear to belong to a single NNE-trending dyke extending ~40 km from Western Channel to Echo Bay (Fig. 2). These sites yielded the shallower inclination. Two sites from a separate NNE-trending dyke yielded a distinct (steeper) direction. Irving et al. (1972) pointed out that the distinct directions observed in the two dykes indicate that they did not cool at the same time. This suggests that they were emplaced at different times, with primary remanences that record secular variation or a small amount of APW. Because 8 of the 10 sites used to calculate the "dyke" mean come from these two intrusions, it is unlikely that the dyke mean pole averages out secular variation.

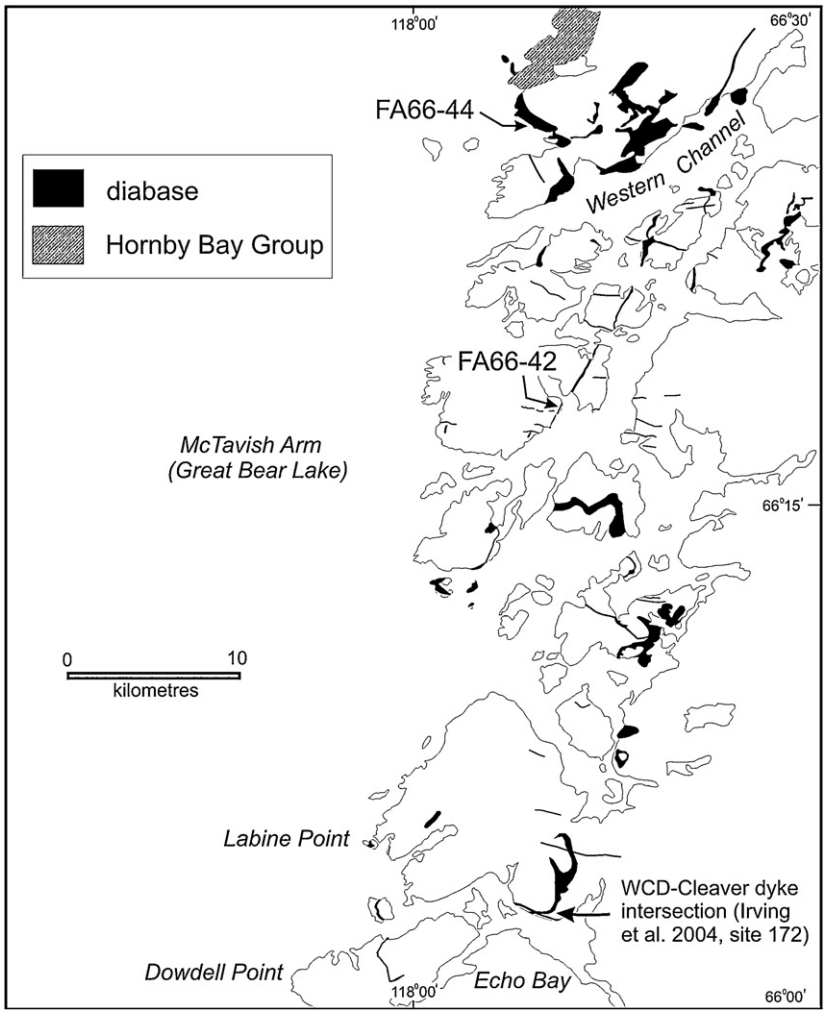


Fig. 2. Geology of the Hornby Bay-Echo Bay area of Great Bear Lake with sampling localities. Diabase is mainly WCD, but WNW-trending dykes in the southeastern portion of the map (and possibly farther north) belong to the Cleaver swarm. Diabase distributions are simplified from Feniak (1952) and Hildebrand (1982).

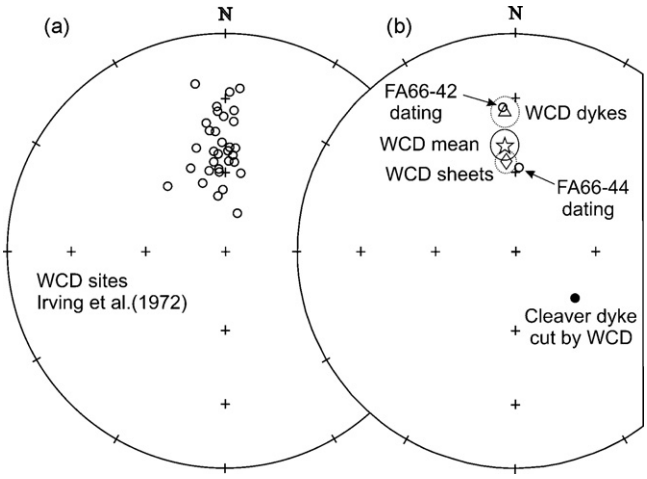


Fig. 3. Paleomagnetic data for the Western Channel Diabase. (a) Site directions from Irving et al. (1972). (b) Paleomagnetic mean directions are shown for sheets (diamond) and dykes (triangle) from the study of Irving et al. (1972). The overall WCD mean direction for all sites in the Irving et al. (1972) study is indicated by a star. Circles show remanence directions of W.F. Fahrig (unpublished archived data) at dated sites FA66-42 and FA66-44 and the direction for a Cleaver dyke cut by WCD near Echo Bay (site 172 of Irving et al. 2004). Open (closed) symbols indicate up (down) directions.

Kidd (1932) suggested that most of the WCD outcrops in flat-lying units may belong to a single sheet or a small number of sheets. The latter interpretation is supported by an examination of the data of Irving et al. (1972). Paleomagnetic directions from sheet sites in local geographical areas fall into distinct clusters. Each cluster could be from a single intrusion that has not averaged out secular variation. This supports the interpretation of the WCD remanence as primary.

Further evidence that the remanence in the WCD is primary was obtained where a WCD sheet cuts across a 1740 Ma Cleaver dyke (Fig. 2; Irving et al., 2004). The Cleaver dyke carries a primary SE down remanence. The sheet carries the N up WCD remanence direction (with the shallower inclination). Although no results are reported from the baked zone of the dyke, these data suggest that there has been no post-WCD overprinting at this locality.

4. Sampling and U–Pb laboratory procedures

The two samples dated in this study were taken from the paleomagnetic collection of W.F. Fahrig, archived at the Geological Survey of Canada, in Ottawa. One sample was chosen from site FA66-42, from which Fahrig (unpublished, archival data) obtained the 'shallower inclination' remanence, and a second from site FA66-44 with the 'steeper inclination' remanence (Fig. 3), in order to

establish if the difference in remanence inclination reflects a significant difference in age of the diabase.

Site FA66-42 is located at 66°18′N, 117°49′W on a SSW-trending dyke or sheet (Fig. 2). This appears to be the same unit along which Irving et al. (1972) obtained 6 stably magnetized sites. The mean paleomagnetic direction at FA66-42 is $D = 355°$, $I = -34°$ ($k = 125$, $\alpha_{95} = 4°$), indistinguishable from the shallower inclination remanence reported by Irving et al. from this intrusion. Site FA66-44 is from a gently dipping sheet at 66°27′N, 117°52′W. The location is along the mainland shore of Great Bear Lake (Fig. 2) near Irving et al.'s (1972) site A40. The mean paleomagnetic direction for FA66-44 is $D = 003°$, $I = -59°$ ($k = 32$, $\alpha_{95} = 10°$). Like the direction for site A40, it falls in the steeper inclination grouping.

Both samples underwent mineral separation and isotopic analysis at the Jack Satterly Geochronology Laboratory at the University of Toronto. In each case, roughly 150 g of sample were processed using conventional crushing and grinding techniques. The resulting coarse powder was then spoon-fed slowly over a shaking water (Wilfley) table, left for several minutes (see Söderlund and Johansson, 2002), and a thin band of heavy minerals was pipetted directly off the table and inspected using a binocular microscope; no heavy liquids or magnetic separation techniques were employed. From each sample, several grains of pale to medium brown, striated baddeleyite blades and blade fragments were recovered. All grains showed highly reflective surfaces and appeared to be free of zircon frostings or overgrowths (discrete zircon grains were also absent from the heavy mineral concentrate). Selected grains were photographed and then analyzed using conventional isotope dilution-thermal ionization mass spectrometry (ID-TIMS) methods. Weight estimates were made from a digital measurement of grain dimensions and the density of baddeleyite.

Baddeleyite grains were washed and loaded into Teflon bombs with concentrated HF along with a mixed ^{205}Pb–^{235}U isotopic tracer solution (Krogh, 1973). Dissolution occurred over 4 days at 195 °C, after which fractions were dried down with phosphoric acid and loaded with silica gel directly onto outgassed rhenium filaments. The isotopic compositions of Pb and U were measured using a single Daly collector with a pulse counting detector on a solid source VG354 mass spectrometer. A detector mass discrimination of 0.053% per atomic mass unit (AMU) and a dead time of 22.8 ns were employed for Daly detector measurements. A thermal source mass discrimination correction of 0.1% per atomic mass unit was applied for both Pb and U. The assigned laboratory blank for U was 0.2 pg, while that for Pb is routinely measured below 1 pg. Error estimates were calculated by propagating known sources of analytical uncertainty for each analysis including within-run ratio variability, uncertainty in the fractionation correction, and uncertainties in the isotopic composition of laboratory blank. Initial corrections were made using an in-house data reduction program. Uncertainties for the ID-TIMS data are given at the 95% (2σ) confidence level. Decay constants used in age calculations are those of Jaffey et al. (1971). Graphical data presentation and quoted ages were generated using the Microsoft Excel Add-in Isoplot/Ex v. 3.00 of Ludwig (2003).

5. U–Pb geochronology—results

Uranium-lead isotopic data for three single grain baddeleyite fractions from each sample are provided in Table 1, and presented graphically together in Fig. 4. Sample weights were estimated to be less than 0.05 μg and the total amount of radiogenic Pb measured in each fraction was less than 45 pg. In most cases, however, the total common Pb (Pb_{com}, Table 1) levels were very low (0.3–0.6 pg); occasional higher levels up to 1.6 pg reflect the difficulty of handling such small grains during loading.

Table 1
U–Pb isotopic data for baddeleyite from Western Channel Diabase.

Analysis no.	Fraction	Weight (μg)	U (ppm)	Th/U	Pb* (pg)	Pb_{com} (pg)	$^{206}Pb/^{204}Pb$ measured	$^{206}Pb/^{238}U$	2σ	$^{207}Pb/^{235}U$	2σ	$^{207}Pb/^{206}Pb$	2σ	$^{207}Pb/^{206}Pb$ Age (Ma)	2σ	% Disc	Rho
FA66-42 MacAlpine Channel																	
MAH6136	Bd-1; 1 wide dbr bl frag	0.05	389	0.106	42.1	1.1	2536	0.277852	0.000586	3.75959	0.01335	0.098135	0.000238	1589.0	4.5	0.6	0.748
MAH7019a	Bd-2; 1 el pbr bl	0.02	168	0.142	9.1	0.4	1381	0.275530	0.001055	3.73704	0.02309	0.098369	0.000420	1593.4	8.0	1.7	0.731
MAH7020a	Bd-3; 1 el pbr bl	0.02	104	0.118	4.5	0.6	525	0.276069	0.002147	3.74551	0.05545	0.098399	0.001107	1594.0	21.1	1.6	0.666
FA66-44 Hornby Bay																	
MAH7021	Bd-1; 1 wide dbr bl frag	0.05	394	0.048	41.7	1.6	1741	0.276738	0.000590	3.75108	0.01677	0.098307	0.000331	1592.3	6.3	1.2	0.692
MAH7022	Bd-2; 1 el pbr bl	0.02	314	0.036	16.5	0.3	3401	0.276309	0.000727	3.74375	0.01318	0.098268	0.000188	1591.5	3.6	1.3	0.845
MAH7023	Bd-3; 1 el pbr bl	0.03	165	0.053	8.7	0.6	1068	0.275843	0.001041	3.74350	0.02728	0.098427	0.000539	1594.5	10.2	1.7	0.679

Notes: el = elongate; br = brown; pbr = pale brown; dbr = dark brown; bl = blade; frag = fragment.
Pbcom – common Pb assuming the isotopic composition of laboratory blank: 206/204-18.221; 207/204-15.612; 208/204-39.360 (errors of 2%).
Th/U calculated from radiogenic $^{208}Pb/^{206}Pb$ ratio and $^{207}Pb/^{206}Pb$ age assuming concordance.
Disc – percent discordance for the given $^{207}Pb/^{206}Pb$ age. Rho – error correlation coefficient. Uranium decay constants are from Jaffey et al. (1971).

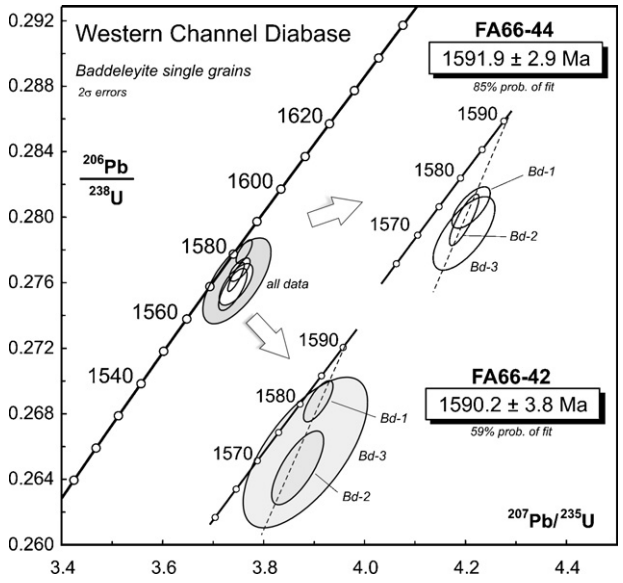


Fig. 4. U–Pb Concordia diagram for Western Channel Diabase samples from sites FA66-42 (shaded ellipses) and FA66-44 (white). Calculated ages are weighted average $^{207}Pb/^{206}Pb$ ages. Dashed lines are regressions through the data, anchored though the origin, for reference.

U–Pb data for both samples are completely overlapping, clustering near concordia with limited spread in Pb/U. Calculated $^{207}Pb/^{206}Pb$ ages for all six fractions range narrowly between 1589.0 and 1594.5 Ma. A weighted average $^{207}Pb/^{206}Pb$ age calculated for single baddeleyite grains from sample FA66-42 is 1590.2 ± 3.8 Ma (59% probability of fit), while an equivalent calculation for the three fractions from FA66-44 is 1591.9 ± 2.9 Ma (85% probability of fit). Both ages agree within error, and are interpreted to represent the age of emplacement and crystallization of the Western Channel Diabase in these localities.

6. Discussion

The U–Pb baddeleyite ages of 1592 ± 3 and 1590 ± 4 Ma reported here provide the first robust dates for the emplacement and crystallization of the Western Channel Diabase. The intrusive relationship of the WCD with host rocks of the upper Hornby Bay Group places a minimum age for deposition of the lower member of the East River Formation at ca. 1590 Ma. This is consistent with recent detrital zircon constraints from clastic members in the upper Hornby Bay Group, suggesting sedimentation post-dated ca. 1620 Ma (Rainbird et al., 2009). The new age is also consistent with the observation that the WCD crosscuts 1740 Ma Cleaver dykes.

Precise dating of the 1.59 Ga WCD and its paleopole has important implications for the drift of Laurentia in the early Mesoproterozoic, as well as for paleocontinental reconstructions that involve Laurentia, Baltica and the Gawler craton of Australia. As discussed above and in Irving et al. (1972, 2004), the WCD remanence is interpreted to be primary. The U–Pb baddeleyite ages obtained from two sites in this study are similar and indicate that the WCD remanence was acquired at ca. 1.59 Ga. Therefore the difference between shallower and steeper remanence directions does not reflect an age difference between the corresponding intrusions of more than 10 m.y., the maximum allowed by the 2σ age uncertainties. We suggest that this difference may be due to secular variation and (or) a small amount of APW.

As discussed earlier, most of the "dyke" sites used by Irving et al. (1972) to calculate a "dyke" mean direction may come from only two separate intrusions, so that it is unlikely that secular variation is averaged out. We conclude that, as the sheet and dyke sites give similar emplacement ages, the most reliable estimate of the WCD mean direction is obtained using both sheet and dyke data, and that the WCD paleopole from these intrusions represents a key pole (Buchan, 2007) suitable for use in constructing a reliable APWP for Laurentia and in rigorous paleocontinental reconstructions.

6.1. APWP for Laurentia

The WCD paleopole falls midway in time and space between key paleopoles for the 1.74 Ga Cleaver dykes of Wopmay orogen and 1.48–1.42 Ga poles for Elsonian-aged units (St. Francois Mountains, Michikamau, Harp Lake, Mistastin and Laramie-Sherman intrusions) from eastern Laurentia (Fig. 5). With the inclusion of the WCD pole, there are no longer time gaps in the APWP from the late Paleoproterozoic to middle Mesoproterozoic of more than 150 m.y., so that the relative polarity of the poles along this APWP segment is more certain than was previously estimated. The path sweeps west from Central America at 1.74 Ga, through the WCD pole to the central Pacific Ocean at Elsonian time. It then defines a deep loop in the north Pacific between 1.27 and 1.087 Ga. Although key poles are not available in the late Mesoproterozoic, non-key overprint poles from the Grenville orogeny, reasonably well dated by Ar–Ar geochronology, suggest that the track continued south, reaching eastern Australia by ca. 1.00 Ga (see more detailed discussion in Buchan et al., 2000). Based on the paleopoles of Fig. 5, the drift of Laurentia in the late Paleoproterozoic–Mesoproterozoic is illustrated in Fig. 6a.

6.2. Laurentia–Baltica comparison

Buchan et al. (2000) analyzed the record of key paleopoles for Laurentia and Baltica that was available at that time. They identified key Baltica poles of three distinct ages (ca. 1.63, 1.575 and 1.265 Ga) in the latest Paleoproterozoic and the Mesoproterozoic. These poles locate Baltica as shown in Fig. 6b. Note that because of magnetic polarity ambiguity, when compared with Laurentia (Fig. 6a), Baltica can occupy either of the two positions shown at a given time.

Buchan et al. (2000) concluded that the only key pole match in the latest Paleoproterozoic–Mesoproterozoic interval was at ca. 1.265 Ga. Utilizing primary poles from the ca. 1.267 Ga Mackenzie magmatic event of Laurentia and the ca. 1.265 Ga Central Scandinavian Dolerite Group (CSDG) of Baltica, and an evaluation of pre-1.265 Ga geology, they proposed a reconstruction in which Baltica is located adjacent to eastern Greenland. A rather similar reconstruction was originally proposed by Poorter (1976) before precisely dated paleopoles were available. Recent U–Pb baddeleyite dating of the CSDG suggests that it was emplaced over a longer period than previously thought (1.271–1.247 Ga; Söderlund et al., 2006). Some of the paleomagnetic sites that were used to calculate the mean CSDG pole are likely from units now dated at ca. 1.258–1.256 Ga, whereas others are dated at ca. 1.265 and 1.268 Ga. Therefore, the Mackenzie-CSDG reconstruction should now be considered to be based on a comparison of ca. 1.267 Ga Laurentia poles with ca. 1.268–1.256 Ga Baltica poles, rather than with 1.265 Ga Baltica poles.

As discussed above, the Western Channel Diabase paleopole is a key pole, based on earlier paleomagnetic work and the precise ages provided in this study. As such, it can be compared with key poles from Baltica. There is no direct age match, but 1.63 and 1.575 Ga key poles for Baltica (Buchan et al., 2000) are only slightly older and younger, respectively, than the WCD pole. In particular, Subjotnian quartz porphyry dykes with U–Pb zircon ages of 1639 ± 9 and 1617 ± 2 Ma (Neuvonen, 1986) and ~1633 Ma (Törnroos, 1984) have a mean pole at 29°N, 177°E (A_{95} = 5°), based

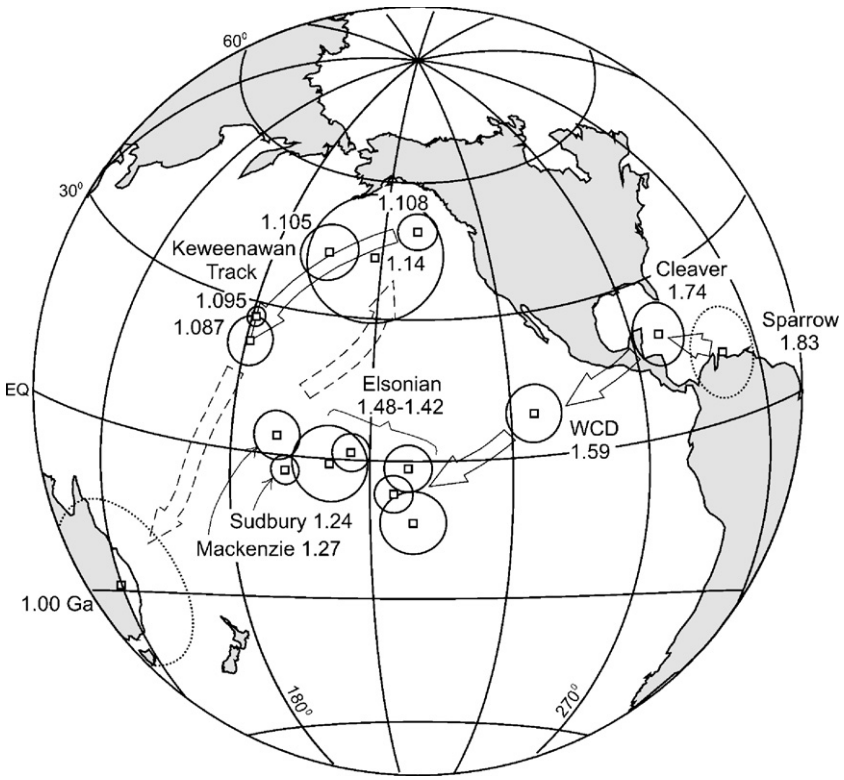


Fig. 5. Rudimentary APWP for Laurentia in the late Paleoproterozoic and Mesoproterozoic based mainly on key paleomagnetic poles. Non-key poles for the 1.83 Ga Sparrow dykes and the Grenville Province at ca. 1.00 Ga, which are discussed in the text, are also shown (with dashed confidence circles). Poles are from Buchan et al. (2000), Meert and Stuckey (2002), Irving et al. (2004) and this study. Arrows indicate the time sequence of poles, rather than a detailed APWP, which may be more complicated. Pole ages are in Ma.

on paleomagnetic data of Neuvonen (1986) and Mertanen and Pesonen (1995). Åland quartz porphyry dykes with U–Pb ages of 1576 ± 13 and 1571 ± 20 Ma (Suominen, 1991) have a pole at 12°N, 182°E (A_{95} = 7°) (Pesonen and Nuevonen, 1981).

In Fig. 7(a and b), we compare the position of Laurentia and Baltica at 1.265 Ga (following Buchan et al., 2000, and references therein) and at 1.630–1.575 Ga (as first described by Hamilton and Buchan, 2007a) based on key poles. In each case, Lauren-

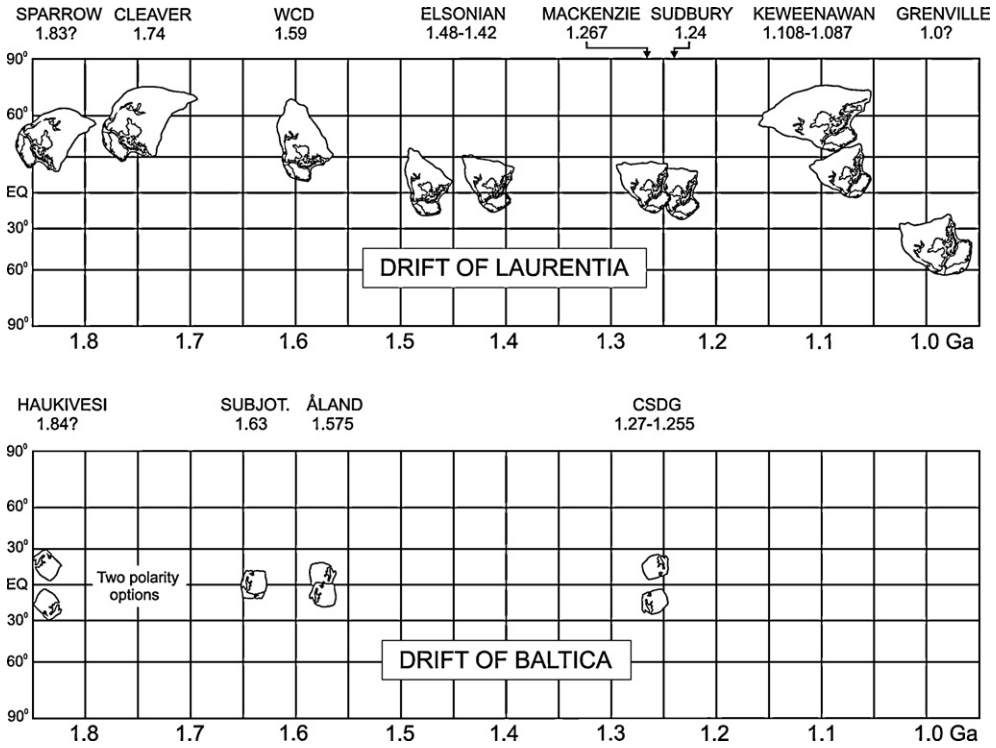


Fig. 6. Late Paleoproterozoic–Mesoproterozoic drift of (a) Laurentia based on paleopoles in Fig. 5, and (b) Baltica based on poles catalogued in Buchan et al. (2000). To avoid overcrowding on the figure, only representative positions are shown in the Elsonian and Keweenawan time intervals. Subjot. = Subjotnian quartz porphyry dykes.

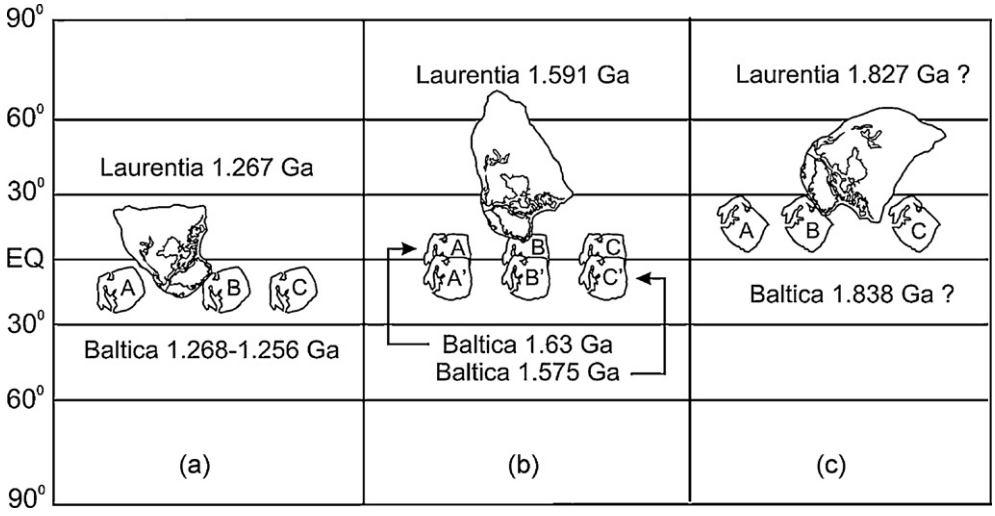


Fig. 7. Comparison of paleolatitude and azimuthal orientation of Laurentia and Baltica based on key paleopoles at (a) ca. 1.27–1.255 Ga and (b) ca. 1.63–1.575 Ga, and based on non-key poles at (c) ca. 1.84–1.83 Ga, as first described by Hamilton and Buchan (2007a). Paleolongitude is not constrained by the paleomagnetic data, so that Baltica can be shifted in longitude relative to Laurentia (e.g., positions A, B or C). Positions labelled B or 'B' represent the fits which are best replicated at all three times. Differences between the three reconstructions, as well as minor overlaps of Baltica and eastern Greenland, may reflect uncertainty in the poles, incomplete averaging out of secular variation and (or) inexact age matches for data from the two cratons. Comparisons (a) and (c) modified from Buchan et al. (2000).

tia is arbitrarily fixed in longitude. Baltica is then permitted to occupy any longitudinal position (e.g., A, B, or C). Only the polarity option for Baltica that permits the closest approach to Laurentia is shown in Fig. 7. The second polarity option (see Fig. 6b) is thought to be unlikely at 1.265 Ga because all Mackenzie and CSDG paleomagnetic data are of a single (presumably the same) polarity.

As discussed in Buchan et al. (2000), if two cratonic blocks drifted together and rotated a significant amount during a given time interval, then relative paleolongitude of the blocks can be determined and their relative position uniquely defined. In the case of the Baltica–Laurentia reconstructions shown in Fig. 7 at 1.63–1.575 and 1.265 Ga, only one approximate fit (labelled B or 'B' in each case) is represented at both times. This places Baltica adjacent to eastern Greenland.

In detail, the similarity between the 'B' reconstructions at ca. 1.63–1.575 and 1.265 Ga, based on key paleopoles, suggests that the two blocks drifted as a unit through the 1.63–1.265 Ga interval. However, it should be noted that the two blocks appear to be rotated ~30° relative to each other in Fig. 7a and b. This may simply reflect uncertainty in the poles, incomplete averaging out of secular variation, and (or) uncertainty in the age match between data for the two blocks. Further refinement of the poles, ages and age matches is needed to resolve this discrepancy. It is interesting to note that a non-key pole comparison at ca. 1.83 Ga (Fig. 7c), which was discussed by Buchan et al. (2000), yields a reconstruction indistinguishable from that at 1.63–1.575 Ga in Fig. 7b. This could indicate that the two blocks drifted as a unit with Baltica adjacent to eastern Greenland from 1.83 Ga. Other paleomagnetists have also looked at the non-key record for Baltica and concluded that the two cratons were likely drifting together at least in the 1.74–1.265 Ga period (Salminen and Pesonen, 2007; Evans and Pisarevsky, 2008). These interpretations need to be tested with key poles.

6.3. Laurentia–Gawler craton comparison

The age of the WCD is identical to a U–Pb titanite age of 1595 ± 5 Ma reported for the Wernecke breccias of the Wernecke and Ogilvie Mountains of the northern Yukon (Thorkelson et al., 2001), and similar to the ca. 1.6 Ga age of magmatism inferred from the inheritance age of zircons in a granite clast from the Coates Lake diatreme in the nearby Mackenzie Mountains (Jefferson and Parrish, 1989) (Fig. 8a). This demonstrates that there is a substantial component of magmatism associated with the 1.59 Ga Wernecke "event" and that this event extends at least 800 km east of the Wernecke Mountains to the Hornby Bay basin and western Wopmay orogen (Fig. 8a).

Thorkelson et al. (2001) proposed that the Wernecke breccias and Olympic Dam breccias of the Gawler craton of southern Australia (Fig. 8b), which have a similar U–Pb zircon age of 1593 ± 7 Ma (Johnson and Cross, 1991), formed a single large (~1000 km diameter) hydrothermal province that may have developed over a mantle plume. This interpretation supported an earlier proposal of such a link made by Bell (1982) before precise ages were available. The Gawler craton is an Archean-Paleoproterozoic craton that extended into Antarctica as the Mawson block, prior to Cretaceous separation of Australia and Antarctica (summarized in Fitzsimons, 2003). The Gawler craton and the Paleoproterozoic–Mesoproterozoic Curnamona Province form parts of the South Australian Craton (SAC) (Fig. 8b). The SAC, North Australian Craton (NAC) and West Australian Craton (WAC) are separated by late Mesoproterozoic mobile belts (Fig. 8b), and are thought by some authors (e.g. Meyers et al., 1996; Betts and Giles, 2006; Wade et al., 2006) to have moved relative to each other during the Mesoproterozoic. The Wernecke and Olympic Dam breccias are coeval with widespread bimodal, though dominantly felsic, Gawler Range Volcanics (GRV) of the Gawler craton whose upper and lower flows have been dated at 1592 ± 3 and 1591 ± 3 Ma, respectively (Fanning et al., 1988). They are also approximately coeval with the Hiltaba granites of the Gawler craton, having ages between ca. 1598 and 1583 Ma (summary in Table 5.3 of Flint et al., 1993), and are marginally older than ca. 1.58 Ga Benagerie volcanics of the Curnamona Province (Fanning et al., 1998). Thorkelson et al.'s (2001) reconstruction of Australia and northwestern Laurentia at 1.59 Ga roughly resembles reconstructions that have been proposed for a variety of time intervals in the late Paleoproterozoic to Neoproterozoic, including the reconstruction of Bell and Jefferson (1987), and the 'SWEAT' reconstruction of Moores (1991) and later authors. However, in the SWEAT reconstruction the SAC is located >1000 km farther south along the western margin of Laurentia than shown by Thorkelson et al.

A comparison of the key 1.59 Ga WCD paleopole for northwestern Laurentia and a similar-aged key pole for the GRV of the Gawler craton would provide a test of the Thorkelson et al. (2001) recon-

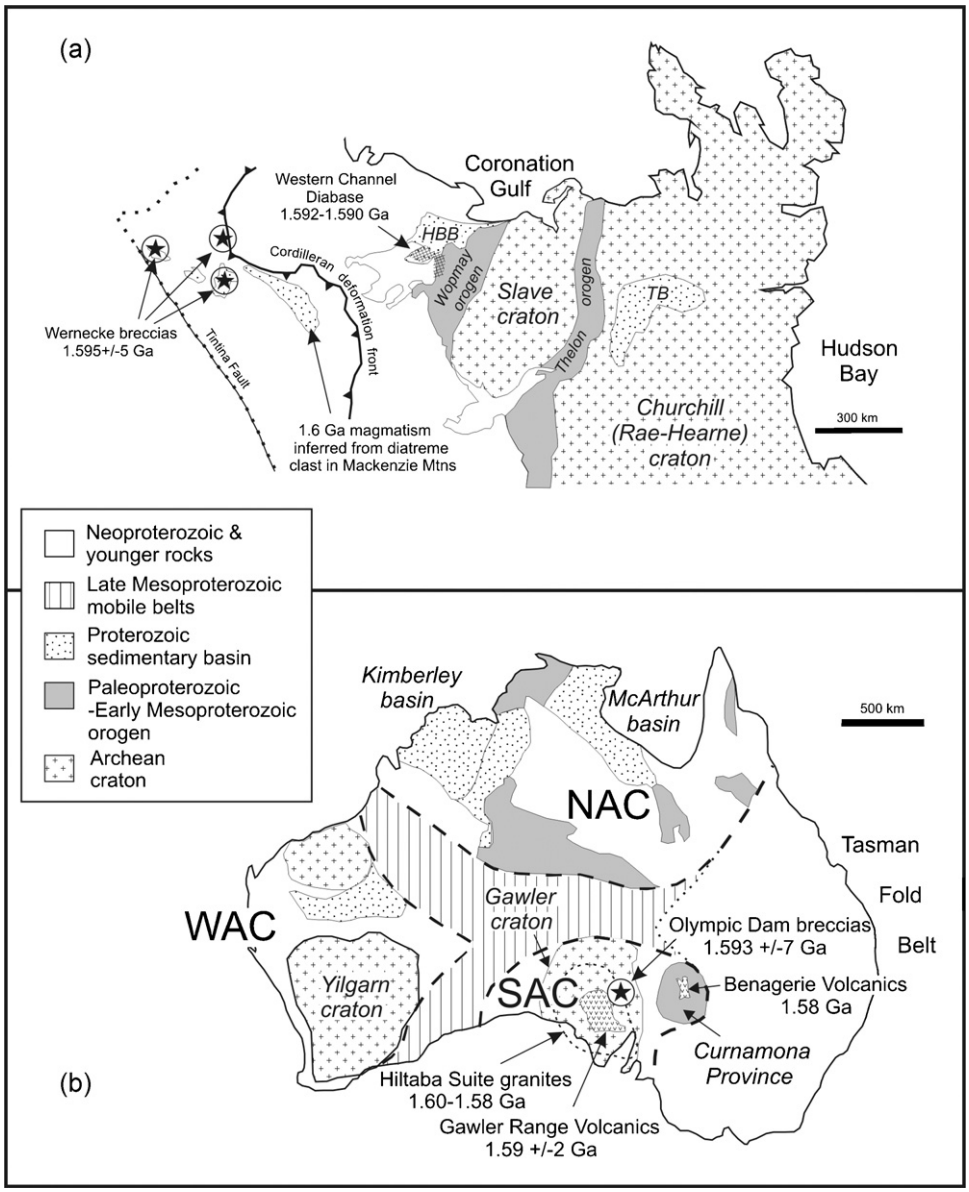


Fig. 8. Ca. 1590 Ma events in (a) northwestern Laurentia and (b) the South Australia Craton. (a) Western limit of the stable craton is after Price and Monger (2003). HBB: Hornby Bay basin; TB: Thelon Basin. (b) The full extent of the Gawler craton and Curnamona Province are shown, including the area beneath Phanerozoic cover rocks. NAC, WAC and SAC represent North, West and South Australia cratons, respectively.

struction. A paleopole has been reported for the GRV, but the age of this pole is open to interpretation. Chamalaun and Dempsey (1978) reported a high-blocking temperature magnetic remanence with a consistent direction from the GRV in the Kokatha area of the Gawler craton. They interpreted the corresponding paleopole as primary but no field test was carried out.

More recently, the age of the GRV paleopole was questioned by Schmidt and Clark (1992) who sampled steeply dipping GRV in a local area about 200 km southeast of Kokatha. They obtained a high-blocking temperature remanence with a direction (with respect to horizontal) that is similar to that reported by Chamalaun and Dempsey (1978). They interpreted the similarity in paleomagnetic directions from flat-lying and steeply dipping flows as a negative fold test demonstrating that the remanence was acquired after folding. They also observed the GRV remanence direction in some other geological units farther south and concluded that there may have been a craton-wide remagnetization event that overprinted the GRV and many other units, perhaps at ca. 1.05 Ga.

Wingate and Evans (2003), on the other hand, favour a primary age for the GRV remanence. As tilting of the lower GRV is thought to have occurred during its emplacement (e.g. Daly et al., 1998), they argued that the volcanic flows investigated in the Schmidt and Clark study may have been magnetized after tilting, either as they cooled from high temperatures or as a result of overprinting by later flows. In either case the remanence would date from 1.59 Ga. In addition, Ar–Ar hornblende ages of 1.61–1.58 Ga from some areas of the southern Gawler craton (Foster and Ehlers, 1998) suggest that the region has not been at sufficiently high temperatures since 1.58 Ga to thermally reset the high-blocking temperature magnetite or low-Ti titanomagnetites that carry the GRV remanence.

There are three possibilities for the age of the GRV remanence: (1) it is primary (i.e. 1.59 Ga in age), (2) it is secondary, but was acquired during the GRV-Hiltaba magmatic event (i.e. 1.60–1.58 Ga), or (3) that it was acquired as a secondary overprint at a significantly later date. The most likely time at which a later overprint might have been acquired is during the 1.57–1.54 Ga

Kararan orogeny associated with a major tectonic event, involving high-grade metamorphism in the northwest and deformation with shear zones in the central portion of the Gawler craton (Daly et al., 1998; Betts and Giles, 2006; Hand, 2007). Regional-scale overprinting of paleomagnetic remanences after 1.40 Ga seems unlikely because of lack of a post-1.40 Ga metamorphic event in the Gawler craton.

If the GRV paleopole was acquired as a metamorphic overprint significantly after 1.59–1.58 Ga (option 3) then it cannot be compared with the 1.59 Ga WCD pole for Laurentia. If it is primary (option 1) or was acquired during the GRV-Hiltaba event as a secondary overprint (option 2), then the position of the Gawler craton (and SAC) at that time can be compared with the 1.59 Ga position of Laurentia based on the WCD paleopole, as described by Hamilton and Buchan (2007b), and as shown in Fig. 9. In this figure, latitude and azimuthal orientation of each craton is fixed by the paleomagnetic data. Laurentia is placed one hemisphere. Because relative polarity of the GRV and WCD poles is not known and paleomagnetic data does not define paleolongitude, the SAC (including the Gawler craton) is shown in both hemispheres and is permitted to occupy any longitudinal position. In other words, each of the positions shown for the SAC is equally possible based solely on the paleomagnetic data at 1.59 Ga.

One of the permitted reconstructions (C) in Fig. 9 places the Gawler craton in close proximity to the Wernecke breccias of the Wernecke and Ogilvie Mountains and the Western Channel Diabase of northwestern Laurentia, at about the location and orientation proposed by Thorkelson et al. (2001). Such a fit would imply that there was a 1.59 Ga magmatic event spanning at least 2000 km on the two blocks, centred approximately where the southeastern cratonic margin of the SAC adjoined the northwestern margin of Laurentia (see enlargement in Fig. 10). If the 1.59 Ga reconstruction of Fig. 10 is valid, it is unclear over what interval of time the SAC and Laurentia may have drifted together, as a number of authors have proposed significantly different reconstructions, based upon paleomagnetic and other criteria, for later intervals in the Proterozoic (summarized in Li et al., 2008 and references therein).

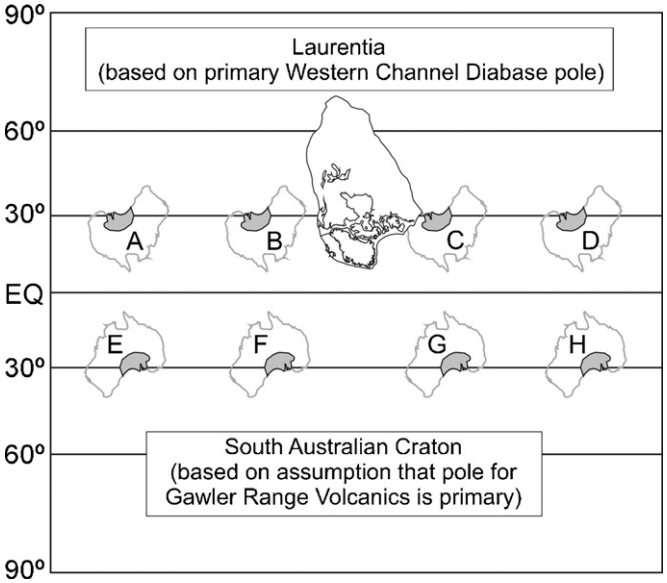


Fig. 9. Reconstruction of Laurentia and South Australian Craton at 1590 Ma, assuming that the Gawler Range volcanic paleomagnetic pole is primary and the Gawler craton and Curnamona Province have not moved relative to one another since that time. Note that the primary nature of the Gawler Range pole is uncertain (see text for discussion). Paleolongitude is not constrained by the paleomagnetic data, so that the South Australian Craton can be shifted in longitude relative to Laurentia (e.g., positions A though D). Because of polarity ambiguity, positions E through H are also possible. Position C represents the closest fit between the SAC and northwestern Laurentia.

If the GRV pole is a late secondary overprint (option 3), then the paleomagnetic reconstruction of Figs. 9 and 10 is not valid, and the apparent association of 1.59 Ga events in the Gawler craton and northwestern Laurentia could be coincidental. Further work is needed to clarify the age of the GRV pole.

There are several important points to note concerning reconstructions of the SAC (or Australia) and northwestern Laurentia.

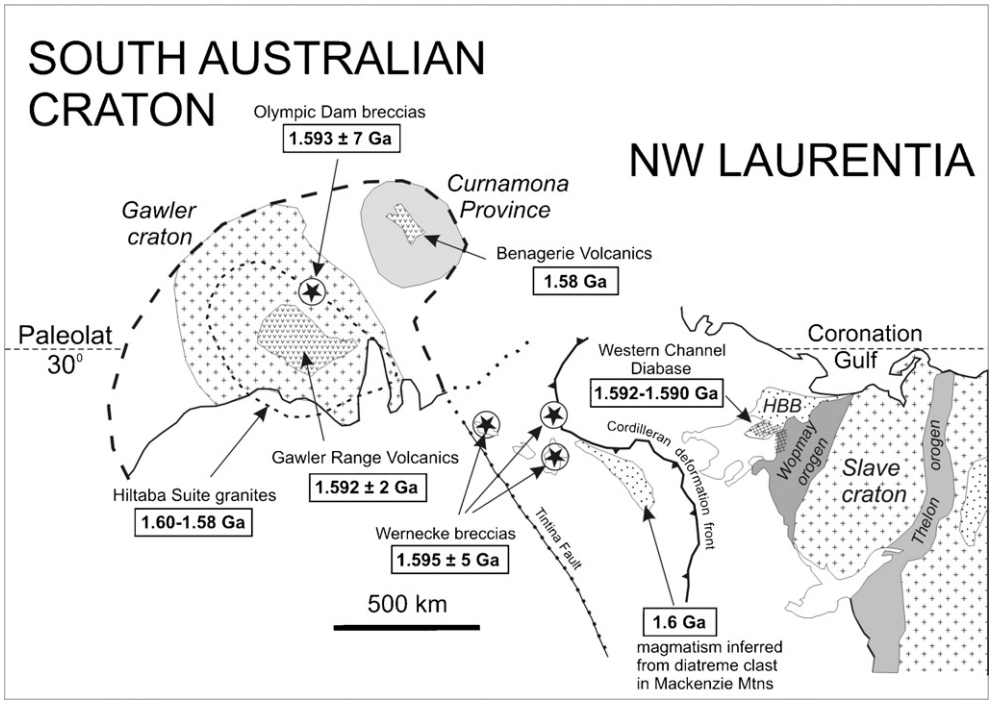


Fig. 10. South Australian Craton reconstructed to northwestern Laurentia based on the closest paleomagnetic fit in Fig. 9 (position C) and the assumption that the Gawler Range paleopole is primary, as first described by Hamilton and Buchan (2007b). HBB: Hornby Bay basin. Symbols as shown in Fig. 8.

(1) To test a reconstruction of the SAC and northwestern Laurentia at 1.59 Ga, a key pole must be determined for the SAC, rather than from elsewhere in Australia, because there is evidence that the SAC has moved relative to the NAC and WAC since that time (Meyers et al., 1996; Betts and Giles, 2006; Wade et al., 2006).
(2) Based solely on the paleomagnetic constraints, the SAC can be positioned farther from the Laurentian margin than shown in Fig. 10. For example, position D (Fig. 9) would permit one or more continental blocks to placed between the SAC and Laurentia. Such a configuration has been proposed by Li et al. (1999) for the supercontinent Rodinia in a much later period (ca. 0.80 Ga), with South China located between Australia and western Laurentia. Of course, placing a block between the SAC and northwestern Laurentia at 1.59 Ga would separate the magmatic/hydrothermal events discussed above.
(3) There are no key paleopoles for the SAC (or Australia) in the 1.59–0.80 Ga period to compare with the Laurentia APWP (Fig. 5) in order to test if the reconstruction of Fig. 9 is tenable after 1.59 Ga. Geological arguments have been made both for and against an Australia-western or northwestern Laurentia connection in the context of a Rodinia supercontinent (e.g., Bell and Jefferson, 1987; Moores, 1991; Rainbird et al., 1996; Karlstrom et al., 1999; Burrett and Berry, 2000; Li et al., 2008). Non-key poles have been used to dispute such a connection at ca. 1.20 Ga (Pisarevsky et al., 2003) and 1.05 Ga (Wingate et al., 2003). Key paleopoles are required to clarify the geographical relationship between the SAC and Laurentia during the Mesoproterozoic.

7. Conclusions

- The Western Channel Diabase (WCD) was emplaced into the Hornby Bay Group and older units of the Great Bear Magmatic Zone at 1592–1590 Ma.
- The new age for the WCD at 1590 Ma represents a minimum age for deposition of the lower member of the East River Formation in the upper part of the Hornby Bay Group.
- The WCD age is similar to that of Wernecke breccias of northwestern Laurentia, indicating a more widespread 1.59 Ga event in northwestern Laurentia than previously recognized.
- The WCD and Wernecke breccia ages are similar to those for the GRV and Olympic Dam breccias of the Gawler craton of southern Australia.
- The WCD paleopole is the first well-dated pole for Laurentia in the 1740–1480 Ma period.
- If the GRV remanence is primary, or if it was acquired during the Hiltaba-GRV event, the paleomagnetic data would permit reconstructions in which the Gawler craton (and SAC) are located adjacent to northwestern Laurentia at 1.59 Ga.
- Baltica was likely located adjacent to eastern Greenland throughout the 1.59–1.265 Ga period, based on evidence from key paleopoles. It may have occupied that position as early as 1.83 Ga.

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